Comparative Evaluation of Keyword, Semantic, and Hybrid Retrieval Strategies for Retrieval-Augmented Generation Systems

An Independent Benchmark-Based Experimental Study

Prepared by: Kothagundla Harshitha

Study type: Independent experimental project / preliminary research report

Evaluation benchmarks: BEIR SciFact and BEIR NFCorpus

Important status note

This document reports the experiments and outputs shared during implementation. It is a preliminary research report, not a peer-reviewed publication. The work evaluates document retrieval components relevant to RAG; it does not implement or evaluate a complete answer-generating RAG pipeline.

Abstract

Retrieval-Augmented Generation (RAG) systems depend on retrieving relevant evidence before a language model generates an answer. This study compares three retrieval strategies: BM25 lexical retrieval, dense semantic retrieval using the Sentence Transformers model *all-MiniLM-L6-v2*, and a hybrid approach that combines rankings using Reciprocal Rank Fusion (RRF). Experiments were conducted using the BEIR SciFact and NFCorpus benchmark test splits. On SciFact, Hybrid RRF achieved NDCG@10 of 0.69217, MAP@10 of 0.64052, Recall@10 of 0.83767, and Precision@10 of 0.09400, outperforming the two individual methods on all four reported metrics. A small weighted-RRF experiment compared BM25/dense weights of 25/75, 50/50, and 75/25; the 50/50 configuration achieved the highest NDCG@10, while 75/25 achieved the highest MAP@10. On NFCorpus, Hybrid RRF achieved the highest reported NDCG@10 and Recall@10, while dense retrieval achieved the highest Precision@10. These results suggest that combining lexical and semantic rankings is promising across the two tested datasets. However, the experiments are preliminary: parameter selection and evaluation were performed on the SciFact test split, no independent held-out validation is yet reported, and the NFCorpus MAP value must be verified from the original CSV because the supplied screenshot partially obscures it. The findings should therefore be interpreted as exploratory rather than definitive.

1. Introduction

Information retrieval is a central component of many RAG architectures. A user query is used to retrieve a small set of potentially relevant passages or documents, which are then supplied to a generative model as evidence. Retrieval quality can influence the relevance and factual grounding of the generated response. Lexical methods such as BM25 rank documents using term-matching statistics, whereas dense retrieval methods encode queries and documents into vector representations and compare their semantic similarity. Sentence-BERT-style models make sentence embeddings suitable for semantic similarity search [2].

A hybrid retriever combines the strengths of lexical and dense approaches. This project investigates whether combining BM25 and dense retrieval rankings improves top-10 retrieval quality on two BEIR datasets. BEIR was designed as a heterogeneous benchmark for evaluating information retrieval systems across different tasks and domains [1].

2. Research Problem

Lexical retrieval can be effective when query and document wording overlap, but it may miss relevant documents expressed using synonyms or semantically related terms. Dense retrieval can capture semantic similarity but may not always preserve exact terminology or domain-specific lexical matches. The research problem is to determine whether combining these complementary ranking signals produces better retrieval effectiveness than either method alone on the selected benchmark tasks.

Research question: How do BM25, dense semantic retrieval, and a hybrid RRF approach compare on SciFact and NFCorpus using standard top-10 information-retrieval metrics?

Objectives: (i) implement BM25 retrieval, (ii) implement dense retrieval using sentence embeddings and FAISS, (iii) combine rankings with RRF, (iv) compare NDCG@10, MAP@10, Recall@10, and Precision@10, and (v) explore a small set of weighted-RRF configurations on SciFact.

3. Related Work

BEIR provides a heterogeneous zero-shot information-retrieval benchmark with datasets spanning tasks such as fact checking and biomedical retrieval [1]. SciFact is a scientific claim verification retrieval dataset, while NFCorpus represents biomedical/nutrition-related information retrieval within BEIR [1, 3, 4].

Sentence-BERT introduces sentence embeddings that can be compared efficiently using cosine similarity, enabling semantic search over collections of texts [2]. Reciprocal Rank Fusion combines ranked lists by accumulating

reciprocal rank contributions; the original work found the method to be a strong rank-fusion approach across several retrieval settings [5].

The present implementation uses these established ideas as baselines and components. It does not propose a new embedding architecture or a new fusion algorithm.

4. Dataset Description

SciFact: The BEIR SciFact test split was loaded through the BEIR data loader. The experiment used the provided corpus, test queries, and relevance judgments (qrels). SciFact is intended for scientific claim retrieval/fact-checking tasks [3].

NFCorpus: The BEIR NFCorpus test split was loaded through the same loader. It provides a second retrieval setting in the biomedical/nutrition information domain [4]. The second benchmark was used to examine whether the observed pattern from SciFact also appeared on a different dataset.

The exact document/query/qrels counts were not included in the saved outputs supplied for this report and are therefore not asserted here. They should be added from the notebook output or dataset metadata before formal submission.

5. Research Methodology

5.1 BM25 lexical retrieval. Corpus documents and queries were lowercased and tokenized using a regular-expression tokenizer. BM25Okapi from the rank-bm25 package scored each document for a query, and the top 10 documents were retained.

5.2 Dense semantic retrieval. Document text was formed by concatenating title and body text. The Sentence Transformers model *all-MiniLM-L6-v2* encoded documents and queries. Embeddings were normalized and indexed using FAISS IndexFlatIP; inner-product search over normalized vectors is equivalent to cosine-similarity ranking. The top 10 documents were returned.

5.3 Hybrid retrieval. BM25 and dense result rankings were combined using Reciprocal Rank Fusion. The RRF score for document d is: RRF(d) = Σr 1 / (k + rankr(d)), with k = 60. The implementation combined the two ranked lists and returned the top 10 documents [5].

5.4 Weighted RRF experiment. On SciFact, the two rankings were combined using weights (0.25, 0.75), (0.50, 0.50), and (0.75, 0.25) for BM25 and dense retrieval, respectively. The selected 50:50 configuration had the highest NDCG@10 among these three tested settings. This is a small exploratory sweep, not a comprehensive optimization.

6. Experimental Setup

The experiments were implemented in Google Colab using Python. The main libraries included BEIR, rank-bm25, sentence-transformers, FAISS, NumPy, Pandas, and Matplotlib. Retrieval depth was set to 10 for the reported benchmark comparison. Evaluation used BEIR's evaluator and the benchmark-provided qrels.

The reported metrics are NDCG@10 (ranking quality with graded relevance and position discounting), MAP@10 (mean average precision over the top 10), Recall@10 (fraction of judged relevant items retrieved in the top 10), and Precision@10 (fraction of the top 10 retrieved items judged relevant). The BEIR evaluator was used to calculate the values.

Reproducibility details that should be recorded before formal submission include the exact BEIR package version, sentence-transformers and FAISS versions, notebook execution date, runtime type, model revision, and full dataset statistics.

7. Results and Discussion

7.1 SciFact results

Method	NDCG@10	MAP@10	Recall@10	Precision@10
BM25	0.65189	0.60697	0.77400	0.08500
Dense	0.64508	0.59593	0.78333	0.08833
Hybrid RRF	0.69217	0.64052	0.83767	0.09400

Hybrid RRF achieved the highest score on all four reported SciFact metrics. Compared with BM25, Hybrid RRF increased NDCG@10 from 0.65189 to 0.69217 and Recall@10 from 0.77400 to 0.83767. Compared with dense retrieval, it increased NDCG@10 from 0.64508 to 0.69217 and MAP@10 from 0.59593 to 0.64052. This supports the observation that the two ranking signals were complementary on this test split.

7.2 SciFact weighted-RRF results

BM25 weight	Dense weight	NDCG@10	MAP@10	Recall@10	Precision@10
0.25	0.75	0.66160	0.61689	0.78333	0.08833
0.50	0.50	0.69217	0.64052	0.83767	0.09400
0.75	0.25	0.67985	0.64310	0.77400	0.08500

The 50:50 setting achieved the highest NDCG@10 (0.69217), Recall@10 (0.83767), and Precision@10 (0.09400) among the three tested configurations. The 75:25 setting achieved the highest MAP@10 (0.64310), slightly above the 50:50 value (0.64052). Therefore, there is no single configuration that maximizes every metric. Also, because these settings were compared on the SciFact test split, the results are exploratory and should not be presented as an unbiased final hyperparameter selection.

7.3 NFCorpus results

Method	NDCG@10	MAP@10*	Recall@10	Precision@10
BM25	0.3061	0.1162	0.1521	0.2180
Dense	0.3159	0.1105	0.1550	0.2433
Hybrid RRF 50:50	0.3381	≈0.1298	0.1667	0.2415

On NFCorpus, Hybrid RRF achieved the highest NDCG@10 (0.3381) and Recall@10 (0.1667) among the three methods. Dense retrieval achieved the highest Precision@10 (0.2433), narrowly above Hybrid RRF (0.2415). The Hybrid MAP@10 value appears to be approximately 0.1298 in the screenshot, but the displayed value is partly obscured; it must be confirmed from the original `nfcorpus_retrieval_comparison.csv` before this report is submitted or cited.

7.4 Overall interpretation

Across the two tested datasets, Hybrid RRF led NDCG@10 and Recall@10. On SciFact it also led MAP@10 and Precision@10. On NFCorpus, dense retrieval had the highest Precision@10. This suggests hybrid fusion may improve ranking coverage and ordering in these experiments, while the best method can depend on the metric and dataset. The absolute metric values should not be compared directly across datasets because the tasks, query distributions, corpus characteristics, and relevance judgments differ.

8. Limitations

1. **Test-set tuning:** SciFact test results were used both to compare methods and to select RRF weights. A separate validation split and untouched final test evaluation are needed for an unbiased estimate.

2. **Limited weight search:** Only three BM25/dense weight pairs were tested. The selected 50:50 setting is best only among these tested options and for the selected metric criteria.

3. **Incomplete NFCorpus MAP record:** The screenshot obscures part of the Hybrid MAP@10 value. The exact value must be checked against the saved CSV or notebook output.

4. **Retrieval-only evaluation:** The study evaluates retrieval metrics, not generated answer factuality, faithfulness, citation correctness, latency, or end-to-end RAG quality.

5. **Single dense model:** Only all-MiniLM-L6-v2 was tested. Other embedding models may produce different rankings.

6. **Implementation details:** Tokenization, title/body concatenation, model revision, package versions, and runtime details should be fully recorded for reproducibility.

7. **Benchmark scope:** Two BEIR datasets do not represent all retrieval domains or production RAG use cases.

8. **Statistical uncertainty:** No confidence intervals, paired significance tests, or per-query error analysis were reported.

9. Conclusion and Future Work

This preliminary study implemented and compared BM25 lexical retrieval, dense semantic retrieval, and RRF-based hybrid retrieval on the BEIR SciFact and NFCorpus test splits. Hybrid RRF achieved the strongest NDCG@10 and Recall@10 results on both datasets. On SciFact, it led all four reported metrics; on NFCorpus, dense retrieval achieved the highest Precision@10. The weighted experiment found that 50:50 produced the highest NDCG@10 among three tested configurations, while 75:25 produced the highest MAP@10.

Future work should use a development split for parameter selection and an untouched test split for final reporting; verify the NFCorpus MAP value from the CSV; expand the weight search on validation data; add additional dense encoders and reranking baselines; perform per-query error analysis and statistical significance testing; measure retrieval latency and memory use; and evaluate a complete RAG pipeline using answer quality and faithfulness metrics.

The current evidence supports a limited conclusion: hybrid ranking was promising on these two benchmark experiments. It does not establish universal superiority or demonstrate that generated answers became more accurate.

10. References

[1] N. Thakur, N. Reimers, A. Rücklé, A. Srivastava, and I. Gurevych, “BEIR: A Heterogeneous Benchmark for Zero-shot Evaluation of Information Retrieval Models,” NeurIPS Datasets and Benchmarks, 2021. https://arxiv.org/abs/2104.08663

[2] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks,” Proceedings of EMNLP-IJCNLP, 2019, pp. 3982–3992. https://aclanthology.org/D19-1410/

[3] BEIR, “SciFact dataset card.” https://huggingface.co/datasets/BeIR/scifact

[4] BEIR, “NFCorpus dataset card.” https://huggingface.co/datasets/BeIR/nfcorpus

[5] G. V. Cormack, C. L. A. Clarke, and S. Büttcher, “Reciprocal Rank Fusion outperforms Condorcet and individual Rank Learning Methods,” Proceedings of SIGIR 2009, pp. 758–759. https://doi.org/10.1145/1571941.1572114

11. Reproducibility and Submission Checklist

Before treating this document as a final academic submission, attach the executed Colab notebook; confirm the NFCorpus Hybrid MAP@10 value from the CSV; record dataset counts and software versions; rerun weight selection using development data only; report final metrics on a held-out test set; add generated graphs from the notebook; and include a repository link once the code is published. Do not describe this report as peer-reviewed or accepted unless that status is independently achieved.